Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Types of Plane Shapes and Their Properties

Plane shapes are shapes that lie flat on a single surface, so they are also called two-dimensional shapes (2-D shapes). They have length and width but no thickness. Common plane shapes include the triangle, rectangle, square, parallelogram, trapezium, circle and other polygons.

Every plane shape can be described by its properties — the facts that are always true about it, such as how many sides it has, whether its sides are equal, whether its angles are equal, how many lines of symmetry it has, and how its diagonals behave. In this section you will learn to recognise each shape, state its properties, and tell how shapes are alike and how they differ.

Look at the three triangles in fig-01. Each tick drawn on a side is a mark that says "these sides are equal in length", and each arc drawn at a corner is a mark that says "these angles are equal".

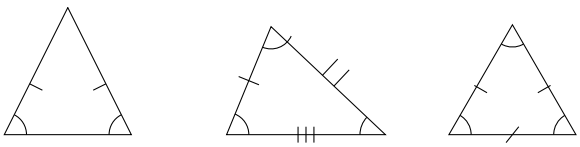
• Isosceles triangle — the two slanted sides carry one tick each, so two adjacent sides are equal. Its base angles (the two angles at the bottom) are equal. It has one line of symmetry.

• Scalene triangle — the three sides carry one, two and three ticks, which means no two sides are equal. So no angle is equal to another. It has no line of symmetry.

• Equilateral triangle — all three sides carry one tick each, so all three sides are equal. That forces all three angles to be equal too, and each is 60°. It has three lines of symmetry.

These three triangles show how the same family of shapes can differ: the isosceles and equilateral both have a line of symmetry, the scalene has none; the equilateral has all sides equal, the isosceles has only two equal, and the scalene has none.

Visual



Isosceles: a single tick on each of the two slanted sides, an arc at each base angle. Scalene: one, two and three ticks on the three sides, an arc at each vertex. Equilateral: a single tick on each of the three sides, an arc at each vertex.

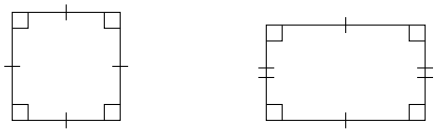
Now compare the square and the rectangle in fig-02. Both are marked with a small right-angle square at each corner, which tells you every angle is a right angle (90°), and both have two pairs of parallel sides. Their difference is in the sides:

• Square — all four sides are equal, and all angles are 90°. It has four lines of symmetry. Its two diagonals cross (bisect) each other at a right angle.

• Rectangle — only the two opposite sides are equal (the longer sides match, the shorter sides match), and all angles are 90°. It has two lines of symmetry. Its diagonals bisect each other but not at a right angle.

So the square and the rectangle agree on their angles (all 90°) and their parallel sides, but they differ on whether the sides are all equal, how many lines of symmetry they have, and how their diagonals cross.

Visual



Squares Rectangles

Square: a right-angle mark at each corner, a single tick on each of the four sides. Rectangle: a right-angle mark at each corner, a single tick on each of the two longer sides and a double tick on each of the two shorter sides.

The parallelogram and the trapezium are both four-sided shapes, but they behave very differently. Look at fig-03: in each shape the single and double arrows on the sides show which pairs of sides are parallel.

• Parallelogram — it has two pairs of opposite sides that are equal and parallel. Opposite angles are equal. It has no line of symmetry. Its diagonals divide it into two equal triangles.

• Trapezium — only one pair of sides is parallel. Its opposite angles are not equal. It has no line of symmetry. Its diagonal divides it into two unequal triangles.

The parallelogram and trapezium share having no line of symmetry, but they differ because the parallelogram has two pairs of parallel sides while the trapezium has only one.

Visual



Parallelogram Trapezium

Parallelogram: a single arrow on each of the two horizontal sides, a double arrow on each of the two slanted sides. Trapezium: a single arrow on each of the two parallel sides, and one diagonal drawn in.

A polygon is a closed shape with three or more straight sides. Polygons are named by their number of sides. A handy pattern connects the number of sides to the number of triangles the polygon can be cut into when you draw diagonals from one vertex to every non-adjacent vertex: the number of triangles is the number of sides minus 2, that is, $n - 2$.

Here is the full list of the common polygons and the triangles each can be divided into:

Triangle 3 sides → $3 - 2 = 1$ triangle
Quadrilateral 4 sides → $4 - 2 = 2$ triangles
Pentagon 5 sides → $5 - 2 = 3$ triangles
Hexagon 6 sides → $6 - 2 = 4$ triangles
Heptagon 7 sides → $7 - 2 = 5$ triangles
Octagon 8 sides → $8 - 2 = 6$ triangles
Nonagon 9 sides → $9 - 2 = 7$ triangles
Decagon 10 sides → $10 - 2 = 8$ triangles
Hendecagon (undecagon) 11 sides → $11 - 2 = 9$ triangles
Dodecagon 12 sides → $12 - 2 = 10$ triangles

Worked example: How many triangles can an octagon be divided into? An octagon has 8 sides, so it divides into $8 - 2 = 6$ triangles.

Practice

1. Name the type of triangle that has no two sides equal, and state how many lines of symmetry it has.
2. A square and a rectangle both have all angles equal (90°). State one property that is true of a square but not of a rectangle.
3. State one similarity and one difference between a parallelogram and a trapezium.
4. How many triangles can a decagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?

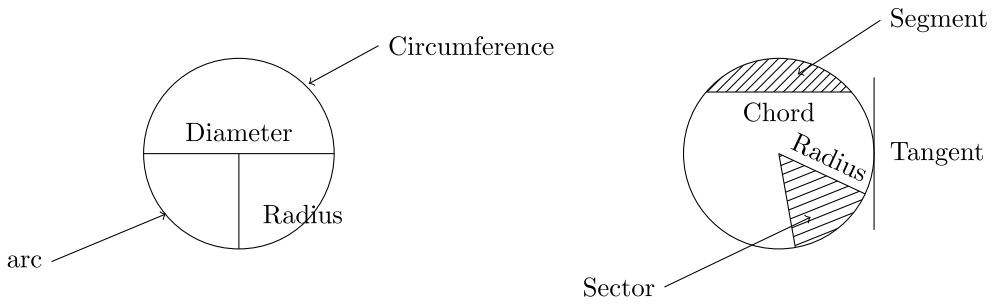
5. Each angle of an equilateral triangle is 60°. Name the triangle in fig-01 that has all three sides equal, and state how many lines of symmetry it has.

Section 2: The Parts of a Circle

A circle is the set of points that are all the same distance from one fixed point, called the centre. A circle has several named parts, and each one earns its name from where it lies or how it is drawn. Look at fig-04, which shows two circles with all the parts labelled, and read each one below.

- **Arc** — a curved part of the circumference. The first circle in fig-04 shows a small curved piece of the rim marked arc: an arc is simply part of the circle's curved edge.
- **Circumference** — the whole curved boundary of the circle. In fig-04 the arrow points to the curved rim of the first circle and labels it the circumference. The arc is part of this boundary.
- **Centre** — the fixed point inside the circle, the one all the points of the circle are the same distance from.
- **Radius** — any straight line joining the centre of the circle to the circumference. In the first circle of fig-04 a radius is drawn straight down from the centre to the rim. The plural of radius is radii.
- **Chord** — any straight line joining two points on the circumference of a circle. In the second circle of fig-04 a chord is drawn across the top, joining two points of the rim.
- **Diameter** — any chord that passes through the centre of the circle. In the first circle the straight line drawn across from one side of the rim to the other, through the centre, is the diameter. Every diameter is a chord, but only a chord that passes through the centre is a diameter.
- **Segment** — the region of a circle cut off by a chord. In the second circle of fig-04, the shaded region above the chord is a segment: it is the part of the circle bounded by a chord and the arc it cuts off.
- **Sector** — the region of a circle bounded by two radii and the arc between them. In the second circle the shaded wedge below the centre is a sector, with its two bounding lines both radii.
- **Tangent** — a straight line that touches the circle at exactly one point. In the second circle the straight line drawn down the right-hand side, touching the rim at a single point, is a tangent.

Visual



Two labelled circles. The first has the diameter drawn across it and a radius from its centre to the circumference, with the arc and the circumference labelled. The second has a chord near the top with the segment above it shaded, a shaded sector below with its right-hand bounding line labelled Radius, and a tangent touching it on the right.

Using the two circles in fig-04, check that you can point to each part and name it.

Take the first circle. The straight line drawn from the centre straight down to the rim is a **radius**, because it joins the centre to the circumference. The straight line drawn across the whole circle through the centre is a **diameter**, because it is a chord that passes through the centre. The curved rim all the way around is the **circumference**, and the small curved piece of it marked with an arrow is an **arc**.

Now take the second circle. The straight line across the top, joining two points of the rim, is a **chord**. The shaded region above that chord is a **segment**, because it is cut off from the circle by the chord. The shaded wedge below the centre, bounded by two radii and the arc between them, is a **sector**. The straight line on the right that just touches the circle at one point is a **tangent**.

Practice

1. A straight line is drawn from the centre of a circle to a point on the circumference. Name this part of the circle.
2. What name do we give to any chord of a circle that passes through the centre?
3. A straight line just touches a circle at exactly one point and does not pass through it. What is this line called?
4. Explain the difference between a chord and a diameter, using the definition of each.
5. Name the shaded region of a circle that is cut off by a chord (as shown above the chord in fig-04).

Section 3: Perimeter of Plane Shapes

Perimeter is the distance all the way round a shape. To find it, you add the lengths of every side that makes up the outside of the shape.

Some shapes repeat the same length several times, so it helps to use a short formula:

- A **square** has four equal sides. If each side is L, the perimeter is $L + L + L + L = 4L$.
- A **rectangle** has two equal long sides (the length L) and two equal short sides (the breadth b). Its perimeter is $L + L + b + b = 2L + 2b = 2(L + b)$.
- A **trapezium** has four different sides, a, b, c and d, so its perimeter is simply $a + b + c + d$.
- A **parallelogram** has two pairs of equal sides. If the base is a and the slanted side is b, its perimeter is $a + a + b + b = 2a + 2b = 2(a + b)$.
- A **regular polygon** (like a regular pentagon or regular hexagon) has all its sides equal. If it has n equal sides of length s, its perimeter is $n \times s$. A regular hexagon has 6 sides, so its perimeter is $6 \times s$.
- A **circle** has no straight sides. Its perimeter is called the **circumference**, and it is found with
  $$C = 2\pi r \quad \text{or} \quad C = \pi d$$
  where r is the radius and $d = 2r$ is the diameter.

Sometimes you are told the perimeter and asked to find a missing side. Work backwards: write the formula, put in what you know, and solve for the unknown length.

Example 1 — Square
A square has sides of 15cm. Its perimeter is
$L + L + L + L = 4L = 4(15) = 60\text{cm}.$

Example 2 — Rectangle
A rectangle has length 10cm and breadth 7cm. Its perimeter is
$L + L + b + b = 2L + 2b = 2(10) + 2(7) = 20 + 14 = 34\text{cm}.$

Example 3 — Finding a missing breadth
The perimeter of a rectangle is 60cm and its length is 17cm. Find the breadth.
$P = 2L + 2b$
$60 = 2(17) + 2b$
$60 = 34 + 2b$
$60 - 34 = 2b$
$2b = 26$
$b = \frac{26}{2} = 13\text{cm}.$

Example 4 — Trapezium
A trapezium has sides 11cm, 9cm, 5cm and 4cm. Its perimeter is
$a + b + c + d = 11 + 9 + 5 + 4 = 29\text{cm}.$

Example 5 — Parallelogram
A parallelogram has base 9cm and slanted side 5cm. Its perimeter is
$2a + 2b = 2(a + b) = 2(9 + 5) = 2(14) = 28\text{cm}.$

Example 6 — Regular hexagon
A regular hexagon has sides of 5.2cm. It has 6 equal sides, so
$\text{perimeter} = 6 \times 5.2 = 31.2\text{cm}.$

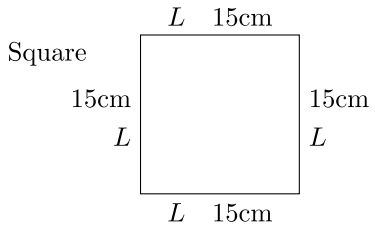
Example 7 — Circle
The perimeter (circumference) of a circle of radius 7cm, using $\pi = \frac{22}{7}$, is
$$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 2 \times 22 = 44\text{cm}.$$

Example 8 — Finding a missing radius
The circumference of a circle is 440cm. Find the radius.
$$C = 2\pi r$$
$$440 = 2 \times \frac{22}{7} \times r$$
$$440 = \frac{44r}{7}$$
Cross-multiplying,
$$440 \times 7 = 44r$$
$$3080 = 44r$$
$$r = \frac{3080}{44} = 70\text{cm}.$$

Example 9 — A shape with a curved part
A shape is a quarter of a circle: two straight sides of 4.9cm each meet at a right angle, and an arc joins their ends. To find its perimeter, add the two straight sides to the length of the curved part.

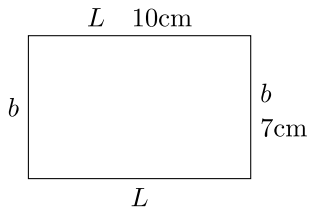
The curved part is one quarter of the circumference of a full circle of radius 4.9cm:
$$\text{curved part} = \frac{1}{4} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 4.9$$
$$= \frac{1}{4} \times \frac{44}{7} \times 4.9 = \frac{11}{7} \times \frac{49}{10} = \frac{77}{10} = 7.7\text{cm}.$$
Now add the two straight sides:
$$\text{perimeter} = 4.9 + 4.9 + 7.7 = 17.5\text{cm}.$$

Visual



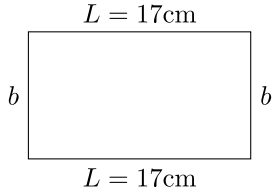
A square, named *Square* at its left, with all four sides marked twice — L and 15cm.

Visual



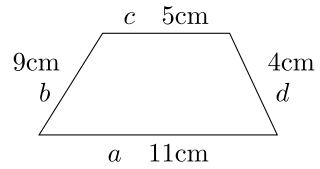
A rectangle with both long sides marked L, the top one also 10cm, and both short sides marked b, the right-hand one also 7cm.

Visual



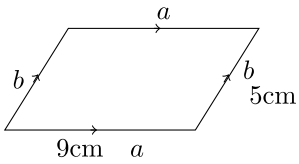
A rectangle with both long sides marked $L = 17$cm and both short sides marked b.

Visual



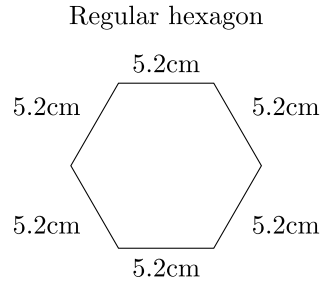
A trapezium with each side lettered and measured: the top c, 5cm; the left b, 9cm; the right d, 4cm; the base a, 11cm.

Visual



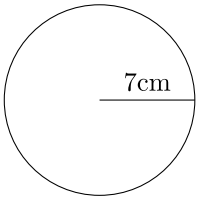
A parallelogram with a single arrow on each of its four sides: the two horizontal ones both lettered a, the lower also measured 9cm, and the two slanted ones both lettered b, the right-hand one also measured 5cm.

Visual



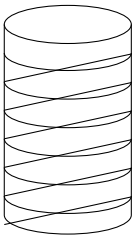
A regular hexagon, named above the drawing, with each of its six sides marked 5.2cm.

Visual



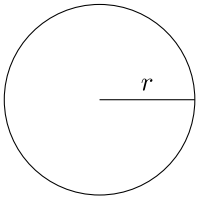
A circle with a radius drawn from the centre to the right of the circumference, marked 7cm.

Visual



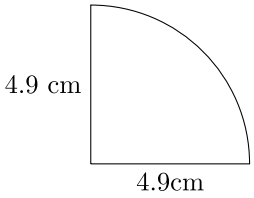
An upright cylinder with a rope wound round it, drawn as a series of turns spiralling down the body from the top face to the base.

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked r.

Visual



A quarter circle: an upright side and a base meeting at a right angle, each marked 4.9cm, with an arc closing them from the top of the upright to the end of the base.

Practice

1. Find the perimeter of a square of sides 8cm.
2. Find the perimeter of a rectangle of length 7cm and breadth 4cm.
3. What is the perimeter of a regular pentagon with sides 5cm?
4. Find the perimeter of a parallelogram whose base is 6cm and slanted sides are 5cm.
5. The circumference of a circle is 22cm. Find the radius of the circle, using $\pi = \frac{22}{7}$.
6. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of the sides of the rectangle in terms of d.
7. Find the perimeter of a regular hexagon with sides 5cm.
8. A rope is wound 50 times round a cylinder of radius 25cm. How long is the rope, using $\pi = 3.14$?

Check your work

Attempt every question before you read this page.

Types of Plane Shapes and Their Properties

1. Scalene triangle; it has no line of symmetry (0 lines).
2. A square has all four sides equal; a rectangle has only its two opposite sides equal. (Also: a square has four lines of symmetry while a rectangle has two, and the diagonals of a square bisect at a right angle while a rectangle's do not.)
3. Similarity: neither has a line of symmetry (both have no line of symmetry). Difference: a parallelogram has two pairs of parallel sides while a trapezium has only one pair; also a parallelogram's opposite angles are equal while a trapezium's are not.
4. 8 triangles ($10 - 2 = 8$).
5. The equilateral triangle; it has three lines of symmetry.

The Parts of a Circle

1. It is a radius.
2. A diameter.
3. A tangent.
4. A chord is any straight line joining two points on the circumference. A diameter is a chord that passes through the centre. So every diameter is a chord, but not every chord is a diameter — only a chord that happens to pass through the centre earns the name diameter.
5. A segment.

Perimeter of Plane Shapes

1. $4 \times 8 = 32\text{cm}$
2. $2(7+4) = 2(11) = 22\text{cm}$
3. $5 \times 5 = 25\text{cm}$
4. $2(6+5) = 2(11) = 22\text{cm}$
5. $r = 3.5\text{cm}$

   $C = 2\pi r$, so $22 = 2 \times \frac{22}{7} \times r = \frac{44r}{7}$. Cross-multiplying gives $22 \times 7 = 44r$, so $154 = 44r$ and $r = \frac{154}{44} = 3.5\text{cm}$.
6. $6d$

   Length $= 2d$, breadth $= d$. Perimeter $= 2(L+b) = 2(2d+d) = 2(3d) = 6d$.
7. $6 \times 5 = 30\text{cm}$
8. 7850cm

   One turn has length equal to the circumference: $2\pi r = 2 \times 3.14 \times 25 = 157\text{cm}$. Fifty turns: $157 \times 50 = 7850\text{cm}$.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-7ae34a68	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____